THE EFFECTS OF INTRODUCING WITHHOLDING ON TAX COMPLIANCE: EVIDENCE FROM PENNSYLVANIA'S LOCAL EARNED INCOME TAX

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This paper examines Act 32, which mandated the introduction of withholding for the local income tax for all employees and the consolidation of a fragmented collection system to one collector per county, effective January 1, 2012, for Pennsylvania local governments. Using a difference-in-differences research design, I find that the act resulted in increased collections of the local income tax by about 9 percent. Results examining the heterogeneous impact of Act 32 show that the effects were larger where fewer residents had been subject to withholding prior to the reform and in counties that experienced a greater consolidation of tax collection.

Keywords: earned income tax, tax compliance, withholding, difference-in-differences

JEL Codes: H26, H71, R51

I. INTRODUCTION

A growing literature in economics emphasizes the importance of information reporting and withholding for tax compliance. Slemrod (2019) notes that the noncompliance rate in the United States varies widely, being as low as 1 percent when there is both information reporting and withholding, such as for wages and salaries, but as high as 63 percent in the absence of third-party information reporting and withholding, as is the case with self-employment income. Withholding by employers plays a particularly key role in ensuring higher compliance, with withholding of individual income and Federal Insurance Contributions Act taxes

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amounting to \$2.7 trillion, or about two-thirds of all federal taxes collected by the Internal Revenue Service (IRS) in 2021 (IRS Data Book, 2021, p. 11).

Driven by a need to obtain higher revenues, governments around the world routinely enact policies that bring more activities under the ambit of information reporting (e.g., Slemrod et al., 2017; Adhikari, Alm, and Harris, 2021) or withholding (e.g., Slemrod and Velayudhan, 2018; Bagchi and Dušek, 2021; Waseem, 2022). However, much of the existing research has examined such policies adopted by national or state governments, with scant attention paid to the experience of local governments. This paper fills that gap in the literature by examining a reform — Act 32 of 2008 — that completely restructured the collection of local earned income taxes (EIT) in the state of Pennsylvania. Although local governments around the state had extensive experience with income taxes, employer withholding was generally not required, and tax noncompliance was a serious issue. In response, Pennsylvania enacted Act 32 into law on July 2, 2008, mandating that by January 1, 2012, all in-state employers must withhold the EIT on behalf of all their employees. Given the lack of third-party information reporting in the pre-Act 32 period, this mandate also meant that for the first time, local tax collectors would start receiving information reports from employers on the wages and salaries earned by their employees. Second, the act streamlined tax collection by establishing 69 regional tax collection districts (TCDs) — roughly one for each county. Last, Act 32 mandated each TCD to set up an information-sharing agreement with the state's Department of Revenue such that it would be relatively easy to detect nonfilers of the local EIT among the population of individuals who had filed a state personal income tax (PIT) return.

The questions I answer in this paper are twofold: First, what were the causal effects of this natural experiment — the state-mandated introduction of withholding and third-party reporting for all employees, the consolidation of tax collection, and setting up information-sharing agreements with the state's Department of Revenue — on collections of the local income tax? Second, and more importantly, to the extent that collections of the income tax increased in the wake of these reforms, what were the underlying mechanisms driving those increases? I construct a panel data set of 466 Pennsylvania school districts and compare collections of the local EIT for Pennsylvania districts with collections of the state PIT *from those same districts* in a difference-in-differences (DID) framework. To address the concern that there could be spillovers from the implementation of Act 32 on state PIT collections, as an alternative, I also compare EIT collections from Pennsylvania with those from Iowa — the only other state where most school districts levy a local income tax. Fortunately, my results are remarkably similar across the two approaches. In either case, using a six-year period around the adoption of Act 32 and school district fixed effects, I find that these reforms led to an increase in EIT collections per capita of about 9 percent. Although my research design addresses the concern that improved economic circumstances may be mechanically driving the higher collections, I also conduct a falsification exercise. I examine collections of the local property tax for the *identical* set of school districts in Pennsylvania and Iowa and confirm that Act 32 did not lead to higher collections of that tax.

Beyond the headline result that EIT collections increase following Act 32, a second and key contribution of this paper is that it sheds light on the underlying mechanisms that underpin the higher collections. I do so through several tests that explore the heterogeneity of these reforms as well as a case study. First, I exploit an institutional feature of the pre-Act 32 regime that had required EIT withholding for employers only when employees lived and worked in the same municipality, whereas after the passage of this reform, employers were required to withhold the tax for all employees, regardless of where they lived. Using variation in the proportion of residents that were employed locally, I find that the effects of these reforms were larger in districts where fewer residents had previously been subject to withholding, pointing to the role of withholding itself in the success of these reforms. Second, I exploit the heterogeneity in the degree of fragmentation of the tax collection system prior to Act 32 and find that school districts located in counties that experienced greater consolidation saw larger increases in collections. Third, I show that although revenues increased more for districts that changed their tax collectors, even districts retaining the same collector experienced significant increases in collections. Fourth and finally, I examine the heterogeneity of response between districts that chose either of the two largest tax collectors, Berkheimer Tax Administrator or Keystone Collections Group, and find that even districts that did not choose one of the "big two" tax collectors experienced an increase in collections. The results of these last two tests suggest that the effects of Act 32 cannot be attributed solely to a selection of collectors who were more efficient; instead, these effects were being driven by the provisions of Act 32, such as the introduction of withholding and the consolidation of tax collection.

In addition to these tests examining the heterogeneous impact of these reforms, I shed light on the mechanisms through a case study involving Lancaster County. Analyzing the change in the number of tax returns filed by residents of Lancaster County, I show that a decline in the rates of nonfiling is one channel that contributes to the rise in collections. Right around the time of adoption of these reforms, there is a sharp increase in the number of local returns filed by taxpayers of about 15 percent, even as the number of state PIT returns filed for those same school districts declines. Furthermore, the average gross income reported on EIT returns *declines* between 2011 and 2012 even as median household income and average gross income reported per state PIT return for the county *increases* slightly. Those facts suggest that changes to the underlying composition of filers following the adoption of Act 32 led to the decrease in average gross income reported, in line with the findings when withholding was introduced for the federal PIT (Holcombe and Gmeiner, 2020) or for the state PIT (Bagchi and Dušek, 2021).

This paper relates to a large and growing literature that has emphasized the role of third-party information reporting in fostering higher levels of tax compliance. Building on earlier work (e.g., Gordon and Li, 2009), Kleven, Kreiner, and Saez

(2016) provide a theoretical framework to show how increasing firm size, accompanied with the use of detailed business records such as accounting books, reduces the likelihood of tax evasion. Empirical evidence to that effect is provided by, among others, Kleven et al. (2011), who report the results of a field experiment in Denmark and find negligible underreporting for income subject to third-party reporting but substantial underreporting for income not subject to such reporting.[1]

The strand of literature this paper most directly contributes to is one that documents the importance of withholding for tax compliance. Tax remittance at source lets revenue agencies collect revenue mechanically even from nonfilers who forfeit any tax their employers have remitted. Among all those who file, the fact that the revenue agency now has a report outlining what the employee's wages are makes it harder for the employee to underreport without triggering an audit. In addition, the presence of withholding affects taxpayer compliance, because individuals who are underwithheld are more likely to underreport or claim deductions. Evidence to that effect is provided by Chang and Schultz (1990), who, using Taxpayer Compliance Measurement Program data, show a correlation between the voluntary compliance rate and whether a balance or refund is due; by Rees-Jones (2018), who confirms that result using observational data for individual taxpayers from the United States; and by White, Harrison, and Harrell (1993) and Vossler, McKee, and Bruner (2021), who substantiate these findings in an experimental setting.

A more recent empirical literature on withholding also finds increases in tax collections through a number of channels: for example, a default payment channel (Waseem, 2022), a change in enforcement perceptions (Brockmeyer and Hernandez, 2022), reduced nonfiling (Bagchi and Dušek, 2021), and other behavioral channels such as reducing frictions in the remittance of taxes (Boning, 2018). One potential concern about drawing inferences from the reform episodes studied in a number of these papers is that they were introduced by governments that directly benefited from higher collections, and it seems plausible that the timing of adoption was influenced by an actual or expected need for revenues to support higher public expenditures.[2] Alternatively, governments might increase the intensity of enforcement simultaneously with these reforms, and data on such changes in enforcement are typically unavailable to researchers.[3] In contrast, the Act 32 reforms studied in this paper were

[1] Another strand of the literature relying on randomized field experiments (e.g., Pomeranz, 2015) or quasi-experiments (e.g., Naritomi, 2019) provides further evidence on the importance of third-party information reporting.

[2] For example, Dušek and Bagchi (2023) note: "Overall, our evidence suggests that the demand for higher government revenues drove the adoption of withholding in the first place and those factors were quantitatively more important in explaining the increase in the size of government than any increase brought about because of adopting a tax system with higher compliance."

[3] For example, in his examination of the impact of bringing the electricity sector under the ambit of the VAT, Waseem (2022) notes: "One other important enforcement event that occurs during my sample period *and may bear on firm compliance* is a nationwide enforcement survey that took place in Pakistan during the period 2000 to 2002. The objective of the survey was to promote documentation of the

designed to improve compliance with the *local* EIT — a source of revenue relied upon by local governments. The date when these reforms would become effective was mandated by the state legislature, with local governments having little discretion on either the mandated date of adoption or the tax base that was constrained to include wages and salaries and self-employment income. Because of these reasons, the endogeneity concerns that might plague prior estimates of the introduction of withholding in other contexts are less relevant here.

In addition, some of the most recent work studying the impact of introducing withholding focus on developing economies such as Costa Rica (Brockmeyer and Hernandez, 2022) and Pakistan (Waseem, 2022), and both do so in the context of the value-added tax (VAT) collected by the national government.[4] Fewer papers examine the effects of introducing withholding for the PIT, where individuals, as opposed to businesses, are responsible for complying with their tax obligations. Moreover, tax collection at the local level entails a different set of dynamics than at higher levels of government. Local taxes pay for public goods that residents more directly consume. In addition, local taxes are chosen by elected politicians who taxpayers may personally know and are collected through a tax system enforced by people who may be neighbors. As a result, the existing literature examining the effects of introducing withholding at the federal or state levels may not be as informative about the importance of introducing withholding for compliance with local taxes.

To the best of my knowledge, the only other paper to examine withholding by local governments is Bagchi (2022), which, like the present paper, analyzes the Act 32 reforms. However, there are a number of important differences between these papers. A key advantage of this paper — which focuses on school districts, whereas Bagchi (2022) focuses on municipalities — is the availability of data. This paper uses a high-quality administrative data set disaggregated by school district from the Pennsylvania Department of Revenue for estimating the tax base for each school district, whereas Bagchi (2022) uses survey data from the American Community Survey (ACS), because it lacks a direct measure of the tax base to which the EIT applies. Next, the results in Bagchi (2022) are obtained using a time-series analysis, and a concern with that approach is that the pre-Act 32 period (2010–2012) includes years heavily affected by the Great Recession; the increased compliance of the local income tax following Act 32 may have simply resulted from an improvement in macroeconomic conditions rather than from the implementation of Act 32.[5] In contrast,

national economy, hoping that it would reduce informality, increase the number of taxpayers, and *bring in more revenue*."

[4] Likewise, Kaçamak, Velayudhan, and Wilking (2023) also examine a policy change — the US Supreme Court's decision in 2018 in *South Dakota v. Wayfair* — that empowered states to pass laws compelling firms (online retailers) to collect sales tax obligations from their consumers.

[5] For instance, the unemployment rate for the three years prior to the introduction of Act 32 (2010–2012) averaged 7.3 percent, whereas the average for the next three years (2013–2015) was more than a full point lower, at 6.0 percent.

this paper benefits from the use of a DID research design, and accordingly, it offers a more credible and cleaner identification approach. Finally and most importantly, this paper provides results that give us a sense of the heterogeneous impact of Act 32 across school districts, and it ties the higher collections following Act 32 to specific provisions of that legislation. This paper also provides a case study that finds a reduction in nonfiling partly drives the increased collections, and it discusses the possibility of consolidating collection of various local taxes in the context of Pennsylvania. In contrast, Bagchi (2022) is silent on the underlying channels leading to the higher tax compliance following Act 32 or these broader questions around consolidation of tax collection.

The rest of the paper is organized as follows. Section II provides institutional background on the local income tax. Section III develops the hypothesis that I test in this paper. Section IV describes the data and my empirical specification. I present the baseline results with the local income tax in Subsection V.A, with the falsification exercise in Subsection V.B. In Section VI, I present results from four triple difference tests that offer us a more nuanced understanding of why collections increased. I present a case study highlighting the role of nonfiling in the pre-Act 32 period in Section VII. Finally, I discuss the desirability of further consolidation of tax collection in Section VIII, and conclude in Section IX.

II. INSTITUTIONAL BACKGROUND

A. State and Local Income Taxes

State and local governments rely heavily on PITs, collecting a combined \$545.1 billion in revenue from these taxes in 2021.[6] To put those numbers in context, that is roughly twice the \$280 billion the federal government collected in corporate income tax revenues in 2021.[7] Thus, PITs are among the most important sources of revenue for state and local governments, and their reliance on these taxes has only grown over time (Bagchi and Dušek, 2021). Of the \$545.1 billion collected by state and local governments in PITs, state governments collected \$503.6 billion, while local governments collected \$41.5 billion. The modest degree of reliance of local governments on individual income taxes stems in part from state rules that limit the ability of those governments to impose an individual income tax of their own. Of the states where local governments levy an income tax, the local tax piggybacks on the state income tax in Indiana, Iowa, Maryland, and New York. Thus, the tax base for the local and the state income tax base coincide in these states, and in addition, taxpayers file their local income tax on the same returns they use for filing the state income tax. In contrast, localities in the remaining states that rely on an income tax — Alabama, Colorado, Delaware, Kentucky, Michigan, Missouri, New

[6] See https://www2.census.gov/programs-surveys/gov-finances/tables/2021/21slsstab1.xlsx.

[7] See https://fred.stlouisfed.org/series/FCTAX.

Jersey, Ohio, Oregon, Pennsylvania, and West Virginia —[8] levy an earnings or payroll tax that is collected and administered separately from the state income tax.[9]

B. The Introduction of Withholding for Personal Income Taxes

At the federal level, a PIT was introduced in 1913 through the adoption of the Sixteenth Amendment.[10] However, for much of the period between 1913 and the onset of World War II, withholding from wages and salaries was not required, and taxpayers remitted their income taxes quarterly, with reconciliation when they filed annual returns.[11] However, in 1943, to pay for the steep expenses associated with the war effort, the federal government introduced withholding of the PIT. Following the introduction of withholding, employers were mandated to withhold a fraction of their workers' paychecks and remit it directly to the government. On their end, taxpayers were required to file income tax returns at the close of the tax year in which they computed the assessed tax. Any difference between the income tax assessed and the amount withheld during the previous year either had to be remitted to the government (if positive) or would be refunded to the taxpayer (if negative).[12]

Thus, employer withholding, as introduced by the federal government in 1944, combined two innovations: third-party information reporting (by which the IRS received reports of employee compensation) and the act of remittance (the employer collecting the tax from their employees and remitting those sums to the agency). With withholding in place, the IRS receives remittances from employers as income is earned, and it does not have to depend solely on employees filing returns at the end of each year. In addition, because the agency receives reports of the compensation paid to employees, it can compare that information with what is reported by a taxpayer on their return and, in the event of a discrepancy, automatically issue a notice of deficiency. The introduction of withholding thus has the potential to increase revenues by making tax evasion more difficult (Kleven, Kreiner, and Saez, 2016).

C. Pennsylvania

Among the roughly 5,000 local governments in the United States imposing an income tax in 2019, 2,964 (or about 60 percent) of those are in the state of

[8] See https://taxfoundation.org/publications/local-income-taxes.

[9] In addition, some localities in Kansas tax interest and dividends but not wages.

[10] An income tax was imposed briefly in 1862 to help pay for Civil War expenses but was repealed in 1872 (https://www.irs.gov/newsroom/historical-highlights-of-the-irs).

[11] Thorndike, Joseph J., "Timelines in Tax History: From 'Class Tax' to 'Mass Tax' during World War II," *Tax Notes*, September 19, 2022, https://www.taxnotes.com/tax-history-project/timelines-tax-history-class-tax-mass-tax-during-world-war-ii/2022/09/16/7f3s2.

[12] The adoption of withholding at the state level happened in a staggered manner between 1948 (Oregon) and 1987 (North Dakota) as laid out in Table 1 in Dušek and Bagchi (2023).

Pennsylvania.[13] This oddity is explained by the fact that states generally limit the ability of their local governments to impose an individual income tax of their own. However, even when states permit localities to impose an income tax, in practice it is often only the largest jurisdictions that do so, such as Detroit (Michigan), Newark (New Jersey), and New York City, whereas in Pennsylvania local governments of all sizes impose this tax. Local income taxes have long been a staple of the state with Philadelphia being the first city in the country to impose an income tax in 1939. Passage of the Local Tax Enabling Act in 1965 authorized other local governments in the state to also impose a tax on wages, salaries, commissions, net profits, or other compensation, and over time the local income tax has grown to become one of the most relied-upon sources of local revenue. In 2017 — the last year for which we have data from the US Census of Governments — this tax raised approximately $5.1 billion, ranking second only to the property tax as a source of revenue. The local income tax is a flat tax applied on all wages, salaries, commissions, net profits, or other compensation and is imposed by municipalities and school districts. The governing body of each jurisdiction that levies the EIT makes its decisions independently of the EIT rate to be levied. However, with limited exceptions, such as those for fiscal distress, the combined EIT rate — across the municipality and the school district — is capped at 1 percent. Consequently, in practice EIT revenues are typically split evenly between the municipality and the school district in which a taxpayer resides.

Despite the state's long experience with the EIT, the collection of this tax was not streamlined. Prior to 2012, employers were required to withhold the EIT only for the small fraction of employees who lived and worked in the same municipality. Second, individual local governments were authorized to select a collector, resulting in a fragmented system with about 560 collectors statewide responsible for collecting taxes in approximately 2,900 jurisdictions (municipalities and school districts). Problems stemming from the lack of withholding and fragmentation included employees not filing taxes with the correct collector, some employees not filing at all, and collectors returning taxes paid by the employer in error rather than forwarding those receipts to the correct agency, leading to an annual estimated revenue loss of $100 million–$237 million (LBFC, 2016, p. 4).[14]

In response, the Pennsylvania state legislature passed, and then-Governor Edward G. Rendell signed into law, Senate Bill 1063, which became Act 32 of 2008. The legislation enacted numerous changes to EIT collection practices, but first and most importantly, the act mandated all businesses within the state to withhold taxes for all employees. Because, on average, fewer than 16 percent of all workers lived and worked in the same municipality, Act 32 had the practical effect of introducing withholding for the EIT, in conjunction with third-party information

[13] See https://taxfoundation.org/local-income-taxes-2019.

[14] Per http://lbfc.legis.state.pa.us/About.cfm, the Legislative Budget and Finance Committee is a bipartisan agency set up to conduct studies and make recommendations aimed at promoting economy in state operations and ensuring that funds are being spent consistent with legislative intent and law.

reporting, when none had existed before.[15] Second, Act 32 required the consolidation of tax collection to one collector for each county (except Allegheny County, which was split into four districts: Central, North, Southeast, and Southwest).[16] The degree of consolidation achieved in practice was even more significant, as two private firms — Berkheimer Tax Administrator and Keystone Collections Group — emerged as collectors of choice and were selected by 48 of the 69 taxing districts (LBFC, 2016, p. 4), thereby reducing the number of collectors from about 560 in the pre-Act 32 period to 18. The duties of tax collectors were spelled out clearly and included the collection, reconciliation, administration, and enforcement of taxes imposed by each political subdivision included in the TCD (53 P.S. § 6924.509(a)(1)), enforcing withholding by employers (53 P.S. § 6924.509(a)(2)), conducting employer and taxpayer audits (53 P.S. § 6924.509(f)), and ensuring that the TCD entered into an information-sharing agreement with the state's Department of Revenue (53 P.S. § 6924.509(g)(1)).[17]

D. Iowa

Iowa is the only state other than Pennsylvania where a majority of school districts levy a local income tax. Like Pennsylvania, the local income tax ranks second only to the property tax as a source of local revenue and, like Pennsylvania, Iowa school districts that impose this tax do not have discretion on the definition of the tax base. However, the collection regime for the local income tax in Iowa is very different from Pennsylvania's. The local income tax is structured as a surtax on top of the Iowa state PIT liability, such that the amount owed as the school district income tax is the product of what an individual owes in her state income tax liability and the surtax rate set by the school district where she resides. Thus, unlike in Pennsylvania, the collection regime of the local income tax is streamlined. Taxpayers report their local tax obligations on the same form as used for their state PIT, and the Iowa Department of Revenue collects this tax and remits it back to school districts. Fortunately for the purposes of the research design of this paper, in which Iowa school districts serve as a control group for Pennsylvania school districts, Iowa's streamlined collection regime has been in place for an extended period of time with no significant changes to either the tax base or tax administration.

[15] In the remainder of this paper, references to the introduction of withholding following Act 32 will encompass the introduction of third-party information reporting in addition to the act of employer remittance of withheld taxes.

[16] Although technically a county, Philadelphia is a city, and there is no independent county government. Philadelphia has its own provisions governing the EIT and was explicitly outside the purview of Act 32.

[17] Act 32 also established strict reporting requirements for tax collectors collecting EIT, and it authorized the Pennsylvania Department of Community and Economic Development to adopt uniform rules, regulations, and forms to be followed and used in each TCD by its designated tax collector.

III. HYPOTHESIS DEVELOPMENT

In terms of the impact of Act 32 on collections of the local income tax, one might expect a positive impact following the mandate to introduce withholding, given the findings from prior research (Holcombe and Gmeiner, 2020; Bagchi and Dušek, 2021) that have documented increased compliance elsewhere when withholding is first introduced.

However, there are at least three reasons to believe that the implementation of Act 32 may have made only a modest difference to tax collections. First, one distinguishing feature of the setting in this paper is that employer withholding had already been in place for the federal income tax since 1944 and for the state PIT since 1971. Pennsylvania residents who are liable for both these taxes knew that revenue agencies at other levels of government had access to third-party information reports from employers even in the absence of withholding by their local governments. With that knowledge, it would perhaps be reasonable to expect that taxpayers were compliant in paying their local income tax obligations faithfully, in which case Act 32 would only result in a difference in *how* the tax obligations were remitted — through self-reporting prior to Act 32 and through employer reporting after Act 32 — rather than *how much* was remitted.

The second reason to suspect a minimal impact of Act 32 on local tax collections is the somewhat limited nature of the tax base and the modest rate at which it applies. The tax applies only to earned income and excludes all nonlabor income, such as that from interest, dividends, and capital gains. Moreover, the combined tax rate is capped at 1 percent, with limited exceptions such as those for jurisdictions experiencing fiscal distress. These caps are binding in practice, which is why the median, as well as the 25th and 75th percentile rates, are all 1 percent. A large body of research (e.g., Clotfelter, 1983; Slemrod, 1985; Alm, Jackson, and McKee, 1992) has documented increasing levels of noncompliance as tax rates go up, and one might anticipate minimal noncompliance with a tax that applies only to earned income and at a rate of only 1 percent.

Finally, a third reason to expect minimal noncompliance prior to Act 32 is because of the level of government imposing this tax and the nature of public goods being funded by those taxes. As is obvious but worth repeating, this paper examines compliance with a local income tax, and surveys of public opinion consistently show that citizens express higher levels of trust in their local governments as opposed to other levels of government.[18] One might well anticipate these higher levels of trust in local government to translate to lower levels of noncompliance. Those expectations of lower noncompliance are strengthened because EIT revenues support spending on local public goods that likely more directly reflect voter preferences and taxpayers' views on their ability to influence government policies plausibly affect noncompliance (e.g., Cowell and Gordon, 1988).

Because of the three reasons offered — the long-standing practice of withholding by higher levels of government, the low rates of the EIT, and the higher levels of trust in local government — we might expect modest levels of noncompliance with the local EIT in Pennsylvania even prior to the passage of Act 32 and therefore anticipate

[18] See https://news.gallup.com/poll/355124/americans-trust-government-remains-low.aspx.

relatively small increases in revenues upon the enactment of these reforms. The subsequent analysis in Section V lets us examine the true magnitude of these reforms on EIT collections.

IV. EMPIRICAL ANALYSIS

A. Construction of Variables and Data Sources

The primary dependent variable used in this paper is the log of local income taxes collected per capita.[19] Constructing this measure requires data on tax collections, which are sourced from the Local Education Agency (School District) Finance Survey (F-33) maintained by the National Center for Education Statistics (NCES). To ensure that increases in collections are not being driven simply by a larger tax base, I control for the size of the tax base using annual reports provided by the Pennsylvania Department of Revenue.[20] These reports include income tax data based on filings of state PIT returns and, importantly for our purposes, are broken out by school district. I extract the data on two components, taxable compensation and net profits, because the local EIT is imposed only on earnings (including self-employment income) but not on sources of nonlabor income such as interest, dividends, and capital gains. These annual reports also provide data on state PIT revenues that are used in the first DID approach in which state PIT collections at the school district level serve as the counterfactual for local EIT collections. I also control for local income tax rates in the regressions, and data on those rates are constructed based on the Municipal Tax Database compiled by the Pennsylvania Department of Community and Economic Development (DCED).[21]

For the alternative estimation in which Iowa's local income tax collections serve as the counterfactual for Pennsylvania's local EIT collections, I source the data on the local tax base and the applicable tax rates from the annual Individual Income Tax School District Reports compiled by the state's Department of Management.[22] I outline the summary statistics and data sources for the control variables in Table 1.

B. Empirical Specification

I use two different control groups for the DID analysis. For the first DID analysis, I exploit the fact that we have data on collections of the state PIT for Pennsylvania

[19] There is no personal exemption or standard deduction for the EIT. See https://tinyurl.com/2pszukuy for an EIT form.

[20] See https://tinyurl.com/2ksucyk7 for an example of the kind of report used.

[21] The tax rates are available at the municipal-year level. Given that Pennsylvania has 500 independent school districts and approximately 2,600 municipalities, most school districts serve more than one municipality. I therefore construct a school district–specific EIT rate by taking the median rate for all municipalities served by a school district.

[22] As noted by the Iowa Department of Revenue, "The surtax base is the statutory amount (Iowa Code Chapter 257.21) to which a surtax percentage may be applied in estimating the amount of surtax that could be collected by a school district. The surtax base is equivalent to line 53 of the 2019 Iowa 1040."

Table 1
Summary Statistics

	Controls at the School District Level			
Variable	Units	Measure	Pennsylvania	Iowa
Median household income	In dollars	Mean	54,457	53,459
		Std. Deviation	15,299	9,638
Percentage of aggregate income from wages and salaries	In percent terms	Mean	71.57	66.56
		Std. Deviation	5.15	6.61
Unemployment rate	In percent terms	Mean	7.13	4.45
		Std. Deviation	2.59	1.96
Percentage of vacant housing units	In percent terms	Mean	12.09	10.68
		Std. Deviation	9.43	5.69
Percentage of the population less than 18 years	In percent terms	Mean	20.97	23.48
		Std. Deviation	2.9	3.05
Percentage of the population older than 65 years	In percent terms	Mean	17.82	18.52
		Std. Deviation	3.1	3.67
Resident local income tax rate	In percent terms	Mean	0.61	0.39
		Std. Deviation	0.26	0.16

Note: Data on median household income, percentage of aggregate income from wages and salaries, unemployment rate, percentage of housing units that are vacant, and percentage of the population that is less than 18 years and older than 65 years are from the five-year ACS. I use six ACSs starting with the 2007–2011 five-year ACS and ending with the 2012–2016 five-year ACS. For instance, data from the 2010–2014 five-year ACS are used for 2012 as it is centered on that year. The five-year estimates are used because they provide data for all areas, whereas the one-year (three-year) estimate only covers areas with populations more than 65,000 (20,000). Data on the Pennsylvania and Iowa resident local income tax rates are sourced from the Municipal Tax Database compiled by the Pennsylvania DCED and from the Individual Income Tax School District Reports compiled by the Iowa Department of Management, respectively.

school districts over time. Pennsylvania state PIT revenues serve as a useful counterfactual to local EIT collections because the economic shocks affecting the state — whether they be declines in unemployment during the recovery from the Great Recession or the fracking boom (e.g., Cunningham, Gerardi, and Shen, 2021; Latzko, 2023) — likely affected state PIT and local EIT collections similarly. Also, importantly for our purposes, there were no changes to enforcement practices for the state PIT around 2012 when the Act 32 reforms were introduced. The one concern about the use of state PIT collections as a counterfactual is the possibility, however remote, that the Act 32 reforms had spillover effects on collections of the state PIT.

To address that concern, I look for another state where school districts can impose a local income tax and do so in practice such that school districts from that state can serve as a counterfactual. In addition, successfully isolating the impact of Act 32 requires the other state to have not conducted a reform of its own tax collection regime around the time that Pennsylvania enacted Act 32. Only two states come close to meeting both criteria: Iowa and Ohio. However, while fewer than a third of Ohio school districts impose a local income tax, 288 of the 361 districts in Iowa had such a tax as of 2012, thereby making it the *only* state apart from Pennsylvania where a majority of school districts levy a local income tax. Moreover, unlike Pennsylvania, the collection of local income taxes has been streamlined in Iowa, with collection performed by the state's Department of Revenue relying on the same forms filed by individuals for complying with the state PIT. The close-to-universal adoption of the local income tax by Iowa school districts, along with the fact that the state did not make any changes to its local tax collection regime during my sample period and the fact that Iowa and Pennsylvania are not geographically proximate, make the state a viable alternative counterfactual for a DID analysis.

For either case, I construct a six-year event window around 2012, the first year in which Act 32 takes effect, to examine its effects on local income tax collections at the school district level. To ensure that variations in population are not driving the variations in collections, I normalize the local income taxes collected in each school district by its population and then take logs to minimize the influence of outliers. I start off with a parsimonious specification that only controls for the local income tax base and tax rates, along with fine-grained school district fixed effects. In a second set of regressions, I include an extensive set of socioeconomic controls, listed below. Thus, the most complete specification used for the DID analysis is

$$Y_{it} = \alpha + \beta_1 * Post_t + \beta_2 * Post_t * 1(PA\ EIT)_i + \beta_3 * X_{it} + \beta_4 * Z_{it} + \lambda_i + \varepsilon_{it}. \quad (1)$$

Y_{it} is the log of local income tax revenues per capita at the school district level. $Post_t$ is a dummy variable set to 1 for the three years following the implementation of Act 32 (i.e., 2013 through 2015) and 0 for the prior three years, 2010 through 2012.[23] X_{it} includes the log of the income tax base per capita and the log of the local income tax rate. Z_{it} are time-variant controls at the school district level: the log of median household income, percentage of aggregate income from wages and salaries, unemployment rate, percentage of housing units that are vacant, and percentage of the population that is less than 18 years and older than 65 years. λ_i are school district fixed effects, and ε_{it} is the error term. My coefficient of interest is β_2 because that coefficient captures the differential impact of Act 32 on local EIT collections for Pennsylvania school districts relative to either counterfactual — state PIT collections from Pennsylvania or local income tax collections from Iowa. Throughout the paper, I cluster standard errors at the county level because Act 32 required each county to have a single tax collector and, in principle, error terms for all school districts within a

[23] The School District Finance Survey (F-33) data provided by NCES are at the fiscal year level. Accordingly, the increased collections resulting from the implementation of Act 32 on January 1, 2012, would show up primarily in the 2013 fiscal year that runs from July 1, 2012, to June 30, 2013, for both Pennsylvania and Iowa.

given county could be correlated. In addition, this choice results in the most conservative standard errors.[24]

Beyond the DID analysis, I also examine the heterogeneous impact of the reform using a triple differences framework. For example, as described more fully in Subsection VI.B, I examine the extent to which the effect of these Act 32 reforms was influenced by the degree of fragmentation in the pre-Act 32 period and the subsequent consolidation of tax collection as mandated by Act 32. Using a dummy variable, *Consolidation*, which captures whether a school district is located in a county that experienced greater-than-average consolidation or not, the empirical specification used for that regression is[25,26]

$$\begin{aligned} Y_{it} = \alpha + \beta_1 {}^*Post_t + \beta_2 {}^*Post_t {}^*1(PA\ EIT)_i + \beta_3 {}^*Post_t {}^*Consolidation_i \\ + \beta_4 {}^*Post_t {}^*1(PA\ EIT)_i {}^*Consolidation_i + \beta_5 {}^*X_{it} + \beta_6 {}^*Z_{it} + \lambda_i + \varepsilon_{it}. \end{aligned} \tag{2}$$

The coefficients of interest in this case are β_2 and β_4, where β_2 measures the impact of the Act 32 reforms in districts that experienced less-than-average consolidation, while β_4 measures the incremental impact of having experienced more-than-average consolidation of tax collection.

V. RESULTS FROM THE BASELINE SPECIFICATION AND THE FALSIFICATION TEST

Before presenting the regression results, a visual representation of the underlying data is helpful. Figure 1 plots the revenues for the local income tax for Pennsylvania and Iowa as well as the state PIT collections from Pennsylvania. The figure shows a sharp rise in EIT collections for Pennsylvania districts right at the time Act 32 is implemented, with no sharp break in trend for Pennsylvania state PIT collections. Examining the raw data, I find that while average local EIT collections increase by about 17 percent, from \$113.13 per person to \$132.35 per person, around the introduction of Act 32, the corresponding increase for state PIT collections from those same districts is about 10 percent.[27] Furthermore, in contrast to the steady increase for state PIT collections in Pennsylvania, local tax collections for Iowa districts oscillate without any predictable pattern during this period. Also upon examining the raw data, I find the increase in Iowa's local tax collections to be smaller: an increase of 8 percent relative to the 17 percent increase for Pennsylvania's EIT

[24] All my results are robust to clustering standard errors at the school district level, and they are available on request.

[25] Pennsylvania state PIT collections serve as the counterfactual for this specification.

[26] The interaction term, $1(PA\ EIT)_i {}^*Consolidation_i$, drops out from the specification with school district FE included.

[27] Excluding one outlier (Bryn Athyn SD, 2015), the post-Act 32 average PIT collections are \$792.41, in which case the mean PIT increase = 8.6 percent. Thus, excluding the one outlier, the relative increase in EIT = 17.0 − 8.6 = 8.4 percent.

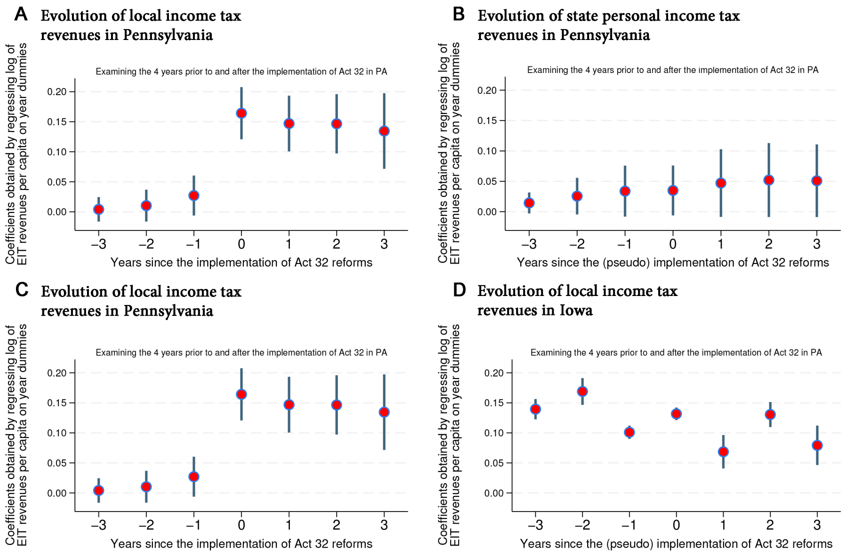

Figure 1. Plot of revenues from the local income tax for an eight-year event window around the year of adoption of Act 32. Panels A and C are identical, and each plots the coefficients, along with the 90 percent confidence intervals, from a regression of the log of local income tax revenues per capita for Pennsylvania school districts on a set of year dummies, all controls, and school district fixed effects, with standard errors clustered at the county level. Panels B and D replicate the exercise of the left-hand panels, with Panel B examining the log of state PIT revenues per capita for Pennsylvania and Panel D examining the log of local income tax revenues per capita for Iowa. For visual presentation, the income tax revenues per capita were winsorized at the 1 percent level. In addition, to ensure that my results are not driven by underlying differences in sample composition, the graph is drawn using only those school districts that show up in each year of the eight-year window. All four figures are drawn using the "coefplot" package (Jann, 2014).

collections. Thus, regardless of which counterfactual we use, I find suggestive evidence that Act 32 positively affected the collection of local EIT revenues for Pennsylvania school districts. Note, though, that this naive comparison of Pennsylvania and Iowa local tax collections does not control for changes over time in the size of the tax base or other variables like the unemployment rate that might independently influence tax collections. Accordingly, a more informed view of the causal effects of Act 32 on local EIT collections can only be drawn through regressions that control for these factors, and it is to those regression results that I turn to next.

A. Baseline Specification: Difference-in-Differences Analysis

Table 2 presents the results from the two DID analyses that contrasts the evolution of local EIT collections for Pennsylvania school districts first with state PIT collections for those same districts (in Columns 1 and 2) and next with local income tax collections from Iowa (in Columns 3 and 4). Columns 1 and 3 are parsimonious

Table 2

Baseline Specification: Effects of the Introduction of Act 32 on Local Income Tax Collections Using a Six-Year Event Window around the Year of Adoption in a Difference-in-Differences Analysis

	Log of Local Income Tax Revenues per Capita			
Dependent Variable	(1)	(2)	(3)	(4)
Control group	Pennsylvania state PIT		Iowa local income tax	
Post-Act 32 × Pennsylvania EIT	0.0869***	0.0870***	0.0951***	0.0949***
	(0.0117)	(0.0117)	(0.0124)	(0.0129)
Post-Act 32	0.0174***	0.0209***	0.0236***	0.0256***
	(0.00191)	(0.00239)	(0.00742)	(0.00707)
Socioeconomic controls included	N	Y	N	Y
Number of observations	5,592	5,592	4,495	4,495

Note: The dependent variable is the log of local income tax revenues per capita, constructed using the Local Education Agency (School District) Finance Survey (F-33) data maintained by NCES. The regressions are estimated on the set of school districts that have both a local income tax and a property tax in place. The analysis excludes Philadelphia, which was outside the scope of Act 32. All columns include a control for the log of the tax base per capita, the log of the resident tax rate, and school district fixed effects. Socioeconomic controls included in Columns 2 and 4 are the log of median household income, percentage of aggregate income from wages and salaries, unemployment rate, percentage of housing units that are vacant, and percentage of the population that is less than 18 years and older than 65 years. Standard errors clustered by county in parentheses. Asterisks denote significance at the 1% (***) level.

and only include (i) a dummy variable that is set to 1 for the post-Act 32 period and 0 otherwise and (ii) its interaction with being present in the treatment group (Pennsylvania EIT), (iii) the log of the income tax base per capita, (iv) the log of the income tax rate, and (v) school district fixed effects. Columns 2 and 4 add several time-variant controls at the school district level, such as the percentage of aggregate income drawn from wages and salaries and the unemployment rate, that could possibly affect tax revenues.

As the results in Table 2 suggest, the implementation of Act 32 is associated with a large and statistically significant increase in local EIT revenues, regardless of what I use as the counterfactual and whether controls are included or not. The coefficient of interest, "Post-Act 32 × Pennsylvania EIT," is statistically significant at the 1 percent level in all four columns, and it varies in a tight range between 8.7 and 9.5 percent. Those estimates suggest a large and statistically significant increase in tax collections of about 9 percent following the implementation of Act 32. The estimates are somewhat smaller in magnitude than the roughly 13 percent increase in compliance reported in Bagchi (2022), and that difference is perhaps accounted for by the use of a DID analysis in this paper as opposed to a simple time-series analysis in Bagchi (2022). Put differently, using Pennsylvania state PIT collections or Iowa's local income tax collections as a counterfactual for Pennsylvania's local EIT revenues reduces the upward bias on the coefficient of interest that stems from better macroeconomic conditions mechanically driving higher collections.

A fundamental concern in any DID analysis is the possibility that the observed effects are being driven by differences in preexisting trends and the presence of these trends is biasing our estimates. If, for example, employers and individuals had started adopting measures to comply with the provisions of Act 32 soon after its passage but prior to its formal adoption on January 1, 2012, then our estimated coefficient of the impact of Act 32 would be biased downward. Alternatively, if local governments were taking steps to increase EIT revenues in addition to the mandated provisions of Act 32, then our estimate of the effects of Act 32 would be biased upward. To better understand the trajectory of the revenue responses, I plot an event-study graph of the impact of Act 32 on local EIT revenues relative to Pennsylvania state PIT collections in Figure 2 and relative to Iowa local income tax collections in Figure 3.

Figures 2 and 3 are reassuring in that they show the parallel trends assumption holds in our sample; there is no unobservable trend in EIT revenues before the implementation of Act 32, given that the pre-Act 32 coefficients are not statistically different from 0. Moreover, the effect of Act 32 is immediate; EIT revenues for Pennsylvania districts rise sharply by more than 13 percent in the adoption year (in both figures), and although there is a dip in subsequent years, the effect of Act 32 continues to be significantly different from 0 during the subsequent years.

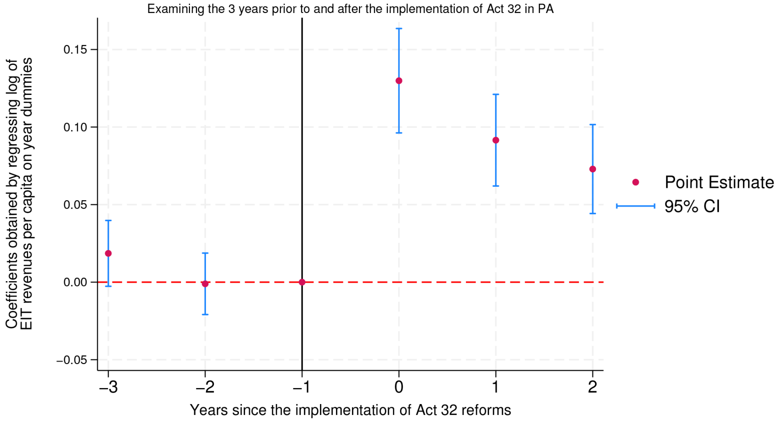

Figure 2. Event-study plot examining the parallel trends assumption for Pennsylvania local EIT collections versus Pennsylvania state PIT collections. The figure plots the coefficients, along with the 95 percent confidence intervals, from a regression of the log of local income tax revenues per capita and state PIT revenues per capita for Pennsylvania school districts on a set of leads and lags, in addition to all controls, and school district fixed effects, with standard errors clustered at the county level. It replicates Column 2 of Table 2 in every respect, with the sole difference being the inclusion of leads and lags (relative to the implementation of Act 32) as opposed to the inclusion of the variables "Post-Act 32" and "Post-Act 32 × Pennsylvania EIT" in generating the estimates in Table 2. This figure is drawn using the "eventdd" package (Clarke and Schythe, 2021).

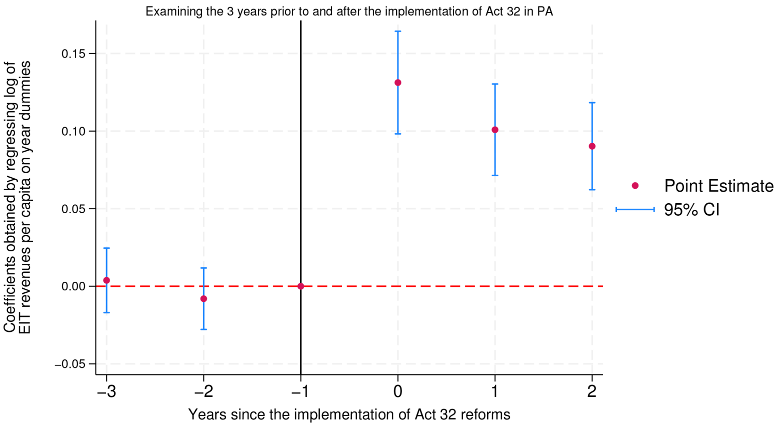

Figure 3. Event-study plot examining the parallel trends assumption for Pennsylvania local EIT collections versus Iowa local income tax collections. The figure plots the coefficients, along with the 95 percent confidence intervals, from a regression of the log of local income tax revenues per capita for Pennsylvania and Iowa school districts on a set of leads and lags, in addition to all controls, and school district fixed effects, with standard errors clustered at the county level. It replicates Column 4 of Table 2 in every respect, with the sole difference being the inclusion of leads and lags (relative to the implementation of Act 32) as opposed to the inclusion of the variables "Post-Act 32" and "Post-Act 32 × Pennsylvania EIT" in generating the estimates in Table 2. This figure is drawn using the "eventdd" package (Clarke and Schythe, 2021).

I subject this finding of increased collections in Pennsylvania school districts to numerous robustness checks that are discussed at length in the Online Appendix (available online). These include varying the length of the window around the adoption of Act 32 to four years (RC1) or eight years (RC2), treating 2006–2008 as the pre-Act 32 period instead of 2010–2012 (RC3), excluding self-employment income from the tax base (RC4), constructing a balanced panel that only includes school districts present in each year of the sample (RC5), and weighting observations by school district population — either current levels (RC6.1) or population as of 2011 (RC6.2). The result that income tax collections increase considerably in Pennsylvania following Act 32 is robust to each of these checks, with the coefficient of interest statistically significant *at the 1 percent level* in 26 of the 28 instances and at the 5 percent level in the remaining 2 instances. I also replicate the results of Bagchi (2022) using a time-series analysis and find estimates larger than those obtained through the DID.[28] Furthermore, I confirm the increases in tax revenues following Act 32 using a matching estimator (Villa, 2016). The results obtained with such an estimator suggest a larger increase in tax revenues of 12.8 percent, compared with the baseline estimate of 9.5 percent in which Iowa school districts serve as the counterfactual for Pennsylvania districts.

[28] All of those results are presented in Table A.1.

B. Falsification Exercise

Although the DID research design addresses the concern that increased collections of the local EIT following Act 32 may have resulted from improved macroeconomic conditions, one may look for additional evidence supporting the claim that the increased EIT collections in Pennsylvania following Act 32 were the causal result of the legislation. It is to provide such evidence that I design a falsification exercise in which I examine revenues from the local property tax for the identical set of Pennsylvania and Iowa school districts, as considered in Columns 3 and 4 of Table 2. This falsification exercise also addresses a concern raised earlier that when governments pass measures designed to enhance compliance with one tax, they might be simultaneously enacting other measures to improve tax compliance across the board, and such measures are often unobserved by the researcher. Here, should that be the case, we ought to find increases in the collections of the local property tax for Pennsylvania beyond increased collections of the local EIT documented above.

One of the advantages of the local property tax is that it is imposed by the same governments that levy the EIT and were affected by the Act 32 reforms. Like local governments elsewhere, this tax is the most important source of revenue in both states, and therefore local governments in these states are likely to be vigilant in ensuring high levels of tax compliance.[29] Also, fortunately for the research design of this paper, there were no changes to the collection regime of property taxes in either state during the sample period, making the falsification exercise useful. Results from such an exercise examining collections of the local property tax are presented in Columns 1 and 2 of Table 3.[30]

In sharp contrast to the earlier results for revenues from the local EIT, I find that the implementation of Act 32 does *not* result in higher collections of the local property tax. Both estimates on the coefficient of interest, "Post-Act 32 $\times$ Pennsylvania," are statistically insignificant, and they change signs depending on the inclusion or exclusion of controls. The results from this falsification exercise lend further support to the claim that the implementation of Act 32 had a causal impact on collections of the local EIT. Had the increase in collections resulted from an across-the-board improvement in macroeconomic conditions, or from general enhancements to tax enforcement, we would have seen increases in collections of the local property tax, but we do not. These results also suggest few spillovers from the passage of Act 32 onto other taxes.

VI. MECHANISMS UNDERLYING HIGHER COLLECTIONS

Although the large increase in local income tax collections following the adoption of Act 32 is unmistakable and holds up to several robustness checks and falsification

[29] See https://taxfoundation.org/state-local-tax-collections.

[30] Event-study plot for this falsification exercise is provided in Figure A.1. It is clear that the property tax does not satisfy the parallel trends assumption, but conditional on the pretrends, there is no discontinuous break in the posttrend coefficients, unlike what we observe for the local income tax.

Table 3

Falsification Exercise: Effects of the Introduction of Act 32 on Collections of the Property Tax Using a Six-Year Event Window around the Year of Adoption in a Difference-in-Differences Analysis

	Log of Property Tax Revenues per Capita	
Dependent Variable	(1)	(2)
Post-Act 32 × Pennsylvania	0.00983	−0.00159
	(0.00924)	(0.00827)
Post-Act 32	0.0593***	0.0493***
	(0.00817)	(0.00795)
Socioeconomic controls included	N	Y
Number of observations	4,495	4,495

Note: The dependent variable is the log of property tax revenues per capita, constructed using the Local Education Agency (School District) Finance Survey (F-33) data maintained by NCES. The analysis includes all Pennsylvania (treatment) and Iowa (control) school districts that have both a local income tax and a property tax in place but excludes Philadelphia, which was outside the scope of Act 32. Both columns include a control for the log of the tax base (measured as the aggregate value of all housing stock in a school district) per capita, the log of the property tax rate (or the millage rate), and school district fixed effects. Socioeconomic controls included in Column 2 are the log of median household income, percentage of aggregate income from wages and salaries, unemployment rate, percentage of housing units that are vacant, and percentage of the population that is less than 18 years and older than 65 years. Standard errors clustered by county in parentheses. Asterisks denote significance at the 1% (***) level.

exercises, pinning down the underlying mechanisms that contribute to this large increase in revenues is challenging. In the following subsections, however, I present results from four additional tests that shed some light on the channels and offer a sense of the heterogeneous effects of Act 32 across districts.

A. Heterogeneity of the Impact Based on Variation in the Extent of Pre-Act 32 Withholding

Although Pennsylvania's local governments have been allowed to impose an income tax since 1965, employer withholding had not been required for collection of the local income tax prior to the passage of Act 32. The only exception was for employees who lived and worked in the same municipality, because employers were required to withhold for such individuals. Pennsylvania municipalities are typically quite small, with the median municipality having fewer than 1,900 residents. As a result, less than a sixth of all residents live and work in the same municipality on average, and those were the individuals who had been subject to withholding prior to Act 32.[31]

[31] The 25th and 75th percentile rates for this variable are 12.0 and 20.2 percent, respectively.

I design a test geared to shed light on the heterogeneous impact of Act 32 by exploiting this institutional feature that had required withholding for a narrow set of employees prior to 2012 and mandated it for all employees thereafter. I partition the overall sample of Pennsylvania school districts based on whether the percentage of residents who lived and worked in the same municipality prior to Act 32 was less than or more than the median.[32] I estimate specification (eq. [2]) but now with an interaction term for school districts where the proportion of residents working outside their municipality of residence pre-Act 32 was more than the median. If the introduction of the employer-withholding mandate had a causal impact on EIT collections, then we would expect to see a positive coefficient on this term when interacted with Post-Act 32 $\times$ Pennsylvania EIT.

Results from this test presented in Column 1 of Table 4 show that although Act 32 led to increases in collections of the EIT across the board of about 7.4 percent, the effects were larger (by about 2.6 percent) in districts where the proportion of residents working outside their municipality of residence was more than the median; that is, for those school districts where withholding had been in place for fewer residents prior to Act 32.[33] This result provides support for the claim that the introduction of withholding played a part in the success of the Act 32 reforms.[34]

B. Heterogeneity of the Impact Based on Reduction in the Number of Collectors

As noted in the institutional context provided in Section II, prior to Act 32, the collection regime in Pennsylvania had been fragmented, with about 560 tax collectors statewide responsible for collecting taxes for approximately 2,900 municipalities and school districts imposing the EIT. This is in sharp contrast to the practice in states like Iowa, where the local income tax is collected centrally by the state's Department of Revenue relying on the same forms used by individuals for filing the state income tax. To address the fragmented collection regime in place, Act 32 essentially mandated a single collector for each county. In practice, this mandate resulted in a substantial reduction for counties with more fragmented systems like Beaver County,

[32] To reduce any concerns about endogenous sorting of residents and to minimize the possible impact of year-to-year fluctuations in the underlying measure, I construct an average over two years *before* Act 32 comes into effect. I observe that while only 12 percent of residents lived and worked in the same municipality in the first subset of school districts on average, 24 percent of residents lived and worked in the same municipality in the second subset of school districts on average.

[33] We can see this result, and results from the other heterogeneity tests, in Figure A.2.

[34] I consider an alternative possibility that it is the size of the school district itself that drives the post-Act 32 increase in EIT revenues, rather than the fraction of residents subject to withholding in the pre-Act 32 period. I estimate several additional regressions to test this hypothesis and find some support in favor thereof. Importantly, while the estimated impact of having fewer residents subject to withholding in the pre-Act 32 period is weaker when I simultaneously control for the size of the district, the results continue to point to an important role for the employer-withholding provisions of Act 32. These additional results are available on request.

Table 4

Heterogeneous Impacts of the Effects of Act 32

	Dependent Variable: Log of Local Income Tax Revenues per Capita						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Post-Act 32 × Pennsylvania EIT × Fewer residents previously subject to withholding	0.0257**				0.0220**	0.0212**	0.0209**
	(0.0109)				(0.00985)	(0.00981)	(0.00972)
Post-Act 32 × Pennsylvania EIT × Greater consolidation in tax collection		0.0480**			0.0403**	0.0388*	0.0344*
		(0.0202)			(0.0192)	(0.0195)	(0.0191)
Post-Act 32 × Pennsylvania EIT × Chose a different collector after Act 32			0.0342***		0.0215*		0.0174*
			(0.0127)		(0.0108)		(0.00973)
Post-Act 32 × Pennsylvania EIT × Chose one of the "big two" tax collectors				0.0455**		0.0338*	0.0313
				(0.0226)		(0.0202)	(0.0193)
Post-Act 32 × Pennsylvania EIT	0.0741***	0.0625***	0.0676***	0.0556***	0.0432***	0.0332*	0.0275
	(0.0103)	(0.0125)	(0.0123)	(0.0198)	(0.0131)	(0.0171)	(0.0180)
Post-Act 32	0.0203***	0.0205***	0.0221***	0.0218***	0.0211***	0.0210***	0.0217***
	(0.00252)	(0.00267)	(0.00237)	(0.00279)	(0.00268)	(0.00301)	(0.00292)
Socioeconomic controls included	Y	Y	Y	Y	Y	Y	Y
Number of observations	5,592	5,592	5,592	5,592	5,592	5,592	5,592

Note: The dependent variable is the log of local income taxes collected per capita, with Pennsylvania state PIT collections per capita serving as the counterfactual. The analysis in this table only examines school districts in Pennsylvania that have both a local income tax and a property tax in place but excludes Philadelphia, which was outside the scope of Act 32. All columns include school district fixed effects and socioeconomic controls: the log of median household income, percentage of aggregate income from wages and salaries, unemployment rate, percentage of housing units that are vacant, and percentage of the population that is less than 18 years and older than 65 years. The specifications also include all relevant interaction terms. For example, Column 1 includes an additional interaction term not reported in this table: Post-Act 32 × Fewer residents previously subject to withholding. Likewise, Columns 2, 3, and 4 include additional terms: Post-Act 32 × Greater consolidation in tax collection; Post-Act 32 × Chose a different collector after Act 32; and Post-Act 32 × Chose one of the "big two" tax collectors, respectively. Complete results, with coefficients on the control variables and the interaction terms, are available from the author on request. Standard errors clustered by county in parentheses. Asterisks denote significance at the 1% (***), 5% (**), and 10% (*) levels.

which went from having 12 distinct tax collectors in 2011 to 1 the following year, while it did not affect the number of collectors in other counties like Clarion County, which had a single collector even prior to Act 32 for all jurisdictions imposing the local income tax.

To examine the degree to which the fragmentation of tax collectors may have played a part in the modest levels of compliance prior to Act 32, I first split the overall sample of Pennsylvania school districts based on whether the reduction in number of collectors experienced by a county between 2010 and 2012 was less than or more than the median.[35] As before, I estimate specification (eq. [2]) with an interaction term for school districts located in counties that experienced a higher-than-average consolidation in tax collection. Note that because Act 32 eliminated discretion on the part of local governments over the number of tax collectors, these differences in the degree of consolidation are driven by ex ante, rather than by ex post, differences. The results I obtain are presented in Column 2 of Table 4, and they demonstrate that districts located in counties that had more fragmented systems experienced additional improvements to EIT collections to the tune of approximately 4.8 percent, resulting from the adoption of a centralized collection regime.

The finding that school districts located in counties that experienced a greater consolidation of tax collection experienced larger increases in EIT revenues following Act 32 raises an interesting question: How were EIT collections influenced by the fragmentation in the pre-Act 32 period? To examine this question, I construct a new dependent variable, the ratio of local EIT to state PIT revenues, and examine how that ratio varies based on number of distinct tax collectors a county had during the pre-Act 32 period. The results from such an analysis are presented in Table 5, and they indicate that during the years 2010 and 2011 when tax collection was fragmented, there was a negative and statistically significant relationship between the ratio of local EIT to state PIT revenues and the number of tax collectors in that county. To get a sense of magnitudes, an increase in the number of tax collectors from 2 (25th percentile of the distribution) to 14 (75th percentile) for the year 2010 would be associated with a drop in the ratio of local EIT to state PIT revenues from 0.187 to 0.144 — a meaningful reduction given that the average level of this ratio is 0.164.[36]

Admittedly, there is an alternative interpretation to the result that counties with more-fragmented tax collection regimes experience larger increases in tax collections

[35] The average number of tax collectors per county in districts that were more fragmented prereform was 17.05 in 2010, and that number goes down to 1.18 in 2012, corresponding to a reduction of 15.87 collectors. In contrast, for counties that were less fragmented prereform, there were only 2.41 tax collectors per county in 2010 and 1.24 collectors in 2012, corresponding to a reduction of 1.18 collectors.

[36] The most common EIT rate is 0.5 percent, whereas the state PIT rate is 3.07 percent. Thus, if the EIT and PIT bases were identical, then the ratio of EIT to PIT revenues would be 0.5/3.07 or 0.163. However, as noted elsewhere, the two bases are not identical. Roughly speaking, the EIT base is 90 percent of the PIT base. Therefore, in a world with perfect compliance, we would expect the EIT to PIT ratio to equal 0.163×0.9 or 0.1467. A higher EIT rate would obviously result in a higher ratio.

Table 5

The Impact of Fragmentation of Tax Collection on Compliance

	(1)	(2)	(3)	(4)
Panel A: Dependent Variable: Ratio of Local EIT Revenues to State PIT Revenues				
Number of EIT tax collectors in that year	−2.201*** (0.651)	−2.268*** (0.672)	−2.031*** (0.633)	−2.126*** (0.639)
Panel B: Dependent Variable: Compliance Rate with the Local Property Tax				
Number of EIT tax collectors in that year	−3.023 (5.570)	−3.567 (5.029)	−4.500 (6.125)	−5.029 (5.391)
Year	2010		2011	
Socioeconomic controls included	N	Y	N	Y
Number of observations	466	466	466	466

Note: The dependent variable in Panel A is the ratio of local EIT revenues to state PIT revenues for a given year (rescaled by multiplying by 100), and in Panel B is the compliance rate with the property tax, defined as the ratio of property tax revenue to the product of the median value of a house in the school district, the number of housing units, and the property tax rate (millage rate; also rescaled by multiplying by 100). Data on local EIT collections are sourced from the Local Education Agency (School District) Finance Survey (F-33) maintained by NCES. Data on state PIT revenues per capita are constructed using annual reports of the Pennsylvania Department of Revenue. Data on the median value of a house and the number of housing units are drawn from the ACS, with data on millage rates drawn from the Municipal Tax Database compiled by the Pennsylvania DCED. Both dependent variables are calculated at the school district level. The independent variable is the log of number of distinct tax collectors for a given county in that year for which I compute the ratio of EIT to PIT revenues or the property tax compliance rate. The socioeconomic controls included in Columns 2 and 4 are the log of median household income, percentage of aggregate income from wages and salaries, unemployment rate, percentage of housing units that are vacant, and percentage of the population that is less than 18 years and older than 65 years. Standard errors clustered by county in parentheses. Asterisks denote significance at the 1% (***) level.

driven by Act 32. Perhaps prereform fragmentation of EIT collection is a stand-in for poor governance in general, and it is poor governance that drives low levels of tax collection in the pre-Act 32 period rather than the fragmentation of tax collection. Although I consider this possibility, I do not find much support in favor thereof. For one, compliance with the property tax — an important source of revenue for these districts — is *uncorrelated* with the level of fragmentation of EIT collection in the pre-Act 32 period, as indicated by the results in Panel B of Table 5. Likewise, I examine another performance measure: test scores on the Pennsylvania System of School Assessment for the school year 2011–2012. Once again, I find no relationship between the extent of fragmentation of EIT collection and measures such as the proportion of students scoring as either (1) advanced/proficient or (2) below basic in either (i) reading or (ii) math, regardless of which measure I use. Thus, based on the evidence, it does not appear that the fragmentation of EIT collection in the pre-Act 32 period is merely a stand-in for poor governance.

C. Heterogeneity of the Impact Based on Whether Tax Collectors Changed Following the Adoption of Act 32

The provision of Act 32 that mandated a single tax collector for each county (except Allegheny) resulted in considerable turnover in the identity of the entities engaged in tax collection. However, not every school district experienced a change in its tax collector. For instance, the two entities that emerged as collectors of choice following the adoption of Act 32, Berkheimer Tax Administrator and Keystone Collections Group, had served about a third of the school districts prior to Act 32. More generally, of the 466 Pennsylvania districts that impose the EIT, 202 (43 percent of the sample) had the same tax collector between 2011 and 2012 (i.e., in the year preceding and the year of adoption of Act 32), whereas the remaining 264 districts (57 percent of the sample) experienced a change in their tax collector between 2011 and 2012. I use that fact to design a test that can help us discern whether improvements in tax collections were driven by a selection of more efficient collectors or whether collections increased even when school districts retained the same tax collector pre- and post-Act 32. I again use a triple interaction approach in which I introduce an interaction term capturing whether a school district switched its tax collector between 2011 and 2012 and reestimate specification (eq. [2]). I present the results thus obtained in Column 3 of Table 4. I see that although districts that changed their collectors did see larger gains (by about 3.4 percent), EIT collections increased by about 6.8 percent even for districts that retained the same tax collector during this period.

D. Heterogeneity of the Impact Based on Whether One of the Two Larger Tax Collectors Was Chosen or Not

Recall from the discussion in Section II that Berkheimer Tax Administrator and Keystone Collections Group were selected by 48 of the 69 taxing districts created as part of Act 32. In this final test, I examine the heterogeneous effects of the Act 32 reforms by looking at whether a school district located in a county choosing either of these "big two" tax collectors had a larger increase in collections or not. I create a dummy variable for whether a school district landed up with either of these two tax collectors and introduce that dummy variable as an additional interaction term in specification (eq. [2]). The results from that test are presented in Column 4 of Table 4, and they indicate that while districts that did not choose one of the "big two" tax collectors experienced an increase in collections of 5.6 percent, districts choosing one of the "big two" tax collectors experienced an additional increase of about 4.5 percent.[37] One reasonable interpretation of these last two tests is that the effects

[37] I find that of the 322 school districts that eventually end up with one of the "big two" tax collectors, 139 had been using their services before (the "stayers"), whereas another 183 had been using a "non-big-two" collector (the "switchers"). Thus, switchers make up 57 percent of the districts landing up with a "big two" tax collector. Further exploration of the data reveals that switchers do, in fact, experience higher average increases in collections relative to the stayers.

of Act 32 cannot be attributed only to a selection of collectors who were more efficient or were among the two biggest tax collectors; instead the provisions of Act 32, such as the mandated introduction of withholding and consolidation of tax collection, led to increased revenues even for school districts that ended up with the same collector or one of the "non-big-two" collectors.

The finding that school districts located in counties that chose one of the "big two" tax collectors experienced larger increases in EIT revenues following Act 32 raises an interesting question: Why were collections higher for these tax collectors compared with the others? At the outset, we can rule out one possible explanation, that districts where these tax collectors were chosen were more affluent and would have had higher collections even otherwise. In my research design, the inference that the "big two" collectors have higher EIT collections is based on the sign and magnitude of a triple interaction term, Post-Act 32 × Pennsylvania EIT × "Has chosen a "big two" tax collector." Thus, the underlying framework in which Pennsylvania state PIT collections serve as the counterfactual for local EIT collections already addresses this concern because more affluent areas would have had higher state PIT collections as well. In addition, each column in Table 4 includes socioeconomic controls such as median household income and incorporates school district fixed effects. Thus, higher EIT collections following the implementation of Act 32 in districts that chose one of the "big two" tax collectors cannot possibly result from those districts being more affluent.

The audit reports that TCDs were required to file following Act 32, available on the DCED website, offer some insights into why districts choosing one of the "big two" tax collectors end up with higher EIT collections. One of the provisions of Act 32 was that the TCD enter into an information-sharing agreement with the state's Department of Revenue such that it would be relatively easy to detect nonfilers of the local EIT among the population of individuals who had filed a state PIT return. Although Act 32 mandated all tax collectors to set up these systems, an audit report issued in August 2017 — *more than five years after Act 32* was introduced — for Berks County's EIT collector finds that it did not have such a system in place. In their report, the auditors note that:

> The Bureau currently does not have an effective system in place to examine the completeness of the tax returns received within taxing jurisdiction. Unless an employer or individual has filed a return in the past, or is filing for a new business license for a taxing district in which the Bureau serves as a collector, limited additional procedures are in place to detect non-filers. The Bureau does receive a file from the Pennsylvania Department of Revenue containing the state income tax returns filed of their taxing jurisdiction. At the present time, the Bureau's software **does not allow** the ability to reconcile those returns with the local tax returns collected by the Bureau.

In contrast, we have evidence (from the audit reports as well as the contracts signed between the taxing districts and the tax collectors) that after the introduction of Act 32, Berkheimer Tax Administrator and Keystone Collections Group were both able to use the information received from the state's Department of Revenue on state PIT filings to effectively pursue nonfilers. In fact, Keystone Collections Group also petitioned the IRS to be designated a "regional income tax agency" so it could set up an information-sharing agreement as part of a Federal/State Exchange Program under I.R.C. § 6103(d)(1).[38]

The audit reports offer another possible explanation for why EIT collections were higher for districts located in counties choosing a "big two" tax collector. The audit reports provide the commission rate charged by each tax collector (as a percent of taxes collected),[39] and I make a note of that for 61 of the 69 taxing districts for 2015, the last year of our sample period (2010–2015).[40] I observe considerable variation in the commission rate charged by various tax collectors, with a minimum of 1 percent for the Allegheny Central Taxing District that includes the City of Pittsburgh to a maximum of 3.17 percent for the Lycoming County Taxing District. More relevant to the issue at hand, I observe a statistically significant, although modest, difference in the rates charged by Berkheimer Tax Administrator and Keystone Collections Group in contrast to the others, with the "big two" tax collectors charging an average commission of 1.68 percent of taxes collected, while the others charge an average commission of 2.02 percent or 0.34 percent higher. Given that the EIT revenue numbers used in our analysis are net of collection costs, these modest differences in the commission rates account for a small, although not trivial, fraction of the difference in EIT revenues between the "big two" collectors and the others.[41] Thus, the implementation of effective information-sharing agreements with the state Department of Revenue following the passage of Act 32 and the lower

[38] See https://www.irs.gov/pub/lanoa/pmta-2022-09.pdf.

[39] 53 Pa. Stat. § 6924.505 requires the tax collection committee to provide for at least one examination for each calendar year of the books, accounts, financial statements, compliance reports, and records of the tax officer by an accountant approved by the tax collection committee. It also requires an audit of all records relating to "the cash basis receipt and disbursement of all public money by the tax officer" and "an analysis of the collection fees charged."

[40] Although taxing districts are required to file audit reports annually with the DCED, in practice, compliance with that mandate is not perfect. I was able to obtain the 2015 audit reports for 56 of the 69 taxing districts, and the commission rate noted in that report is what I use in the subsequent analysis. In a few instances where the 2015 audit reports were unavailable, I used the 2014 or 2016 audit reports instead, to the extent those were available. To be more precise, I use the commission rate provided in the 2014 audit reports for three taxing districts (Allegheny Central, Blair, and Clinton), the rate in the 2016 audit report for one taxing district (Forest County), and in the only instance where the commission rate differs between the 2014 and the 2016 reports (Lycoming County), I use the average of those two rates. In total, I am able to obtain the commission rate charged for 61 of the 69 taxing districts created after the Act 32 reforms. Audit reports and commission rates are unavailable for eight taxing districts (Cameron, Fayette, Fulton, Huntingdon, Juniata, Montour, Perry, and Somerset counties) across the 2014, 2015, and 2016 audit cycles.

[41] A back-of-the-envelope calculation suggests that these differences in commission rates account for 0.34/3.38 or about 10 percent of the difference in EIT collections.

commission rates may partly explain why districts choosing one of the "big two" tax collectors have higher EIT collections.[42]

The finding that school districts being served by one of the "big two" tax collectors experienced greater increases in collections also raises the question of whether there are economies of scale in tax collection. I dig deeper into that issue by returning to the audit reports and plotting the relationship between the commission rate charged by each tax collector and the size of the taxing district in the form of a scatter plot. As Figure 4 indicates, there appears to be clear evidence of economies of scale in tax collection — a conclusion also supported by regression results that are included in Table A.2.

Returning to the heterogeneity tests, one concern is that the various dimensions of heterogeneity could be correlated such that one test may be picking up variation in treatment intensity along another dimension. To address this concern, I introduce multiple dimensions of heterogeneity simultaneously in a regression that interacts treatment status with each dimension. In Column 5 of Table 4, I simultaneously introduce an interaction between the Post-Act 32 × Pennsylvania EIT dummy and (1) a variable indicating whether the proportion of residents subject to withholding prior to Act 32 is less than the median or not, (2) a variable involving whether a county experienced greater-than-average consolidation or not, and (3) whether a different tax collector was chosen or not. The results are illuminating; they indicate all three dimensions of heterogeneity matter independently, as each interaction term is statistically and economically significant. Consolidation of tax collection has the largest incremental impact on EIT revenues of about 4 percent, with the impact of being less exposed to withholding and choosing a different tax collector each equal to about 2.2 percent. Likewise, the results from Column 6 (which has a similar format to Column 5 but considers the choice of the "big two" tax collectors) suggest that each dimension of heterogeneity matters. Once again, the consolidation of tax collection appears to have the greatest incremental impact, about 4 percent, but the impacts of being less exposed to withholding pre-Act 32 and choosing one of the "big two" tax collectors continue to be meaningful. Finally, Column 7 introduces all four

[42] Admittedly, there is an alternative explanation centered on the endogenous selection of tax collectors, in which school districts with the highest noncompliance rates consciously hire a "big two" collector postreform. There are two reasons why I believe this explanation does not have much bite. For one, the selection of a tax collector is made at the level of the TCD, where each school district and municipality gets a say. Given the number of jurisdictions imposing the EIT and the number of TCDs, "on average" each TCD encompasses 7 school districts and 35 municipalities. Thus, no one school district is unilaterally able to select its tax collector postreform based on its own needs. Second, I observe that the compliance rate for school districts that eventually end up with one of the "big two" tax collectors is not significantly different from the set of districts do not end up with a "big two" tax collector. Defining compliance as EIT collected / (EIT rate × EIT tax base), I find that the average pre-Act 32 EIT compliance rate for districts ending up with one of the "big two" collectors is 91.2 percent, whereas for districts ending up with one of the "non-big-two" collectors, the pre-Act 32 average is about 90.6 percent. Thus, although there may be concerns about the endogenous selection of tax collectors, from a practical standpoint, such an alternative explanation does not have much bite here.

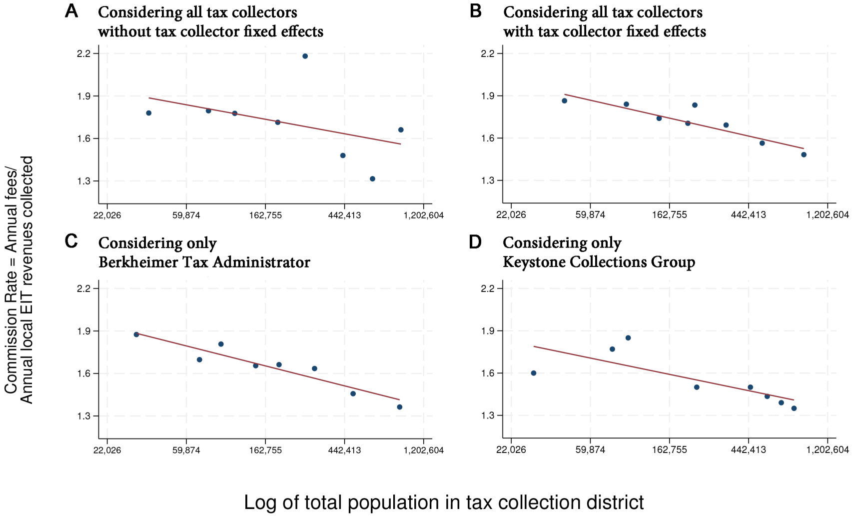

Figure 4. Scatterplot of the relationship between the commission rate charged by tax collectors and the population of the taxing district. This figure shows scatterplots of the relationship between the commission rate charged by tax collectors (as a percentage of EIT collected) and the size of the taxing district, as proxied by the log of population of the taxing district. Data on commission rates are drawn from the audit reports that taxing districts were required to file as a part of Act 32. The underlying data are at the taxing district level and include all 59 districts for which we have data on commission rates from the audit reports filed with the Pennsylvania DCED in either 2014, 2015, or 2016 and where at least one school district imposed the EIT. This figure is created using the "binscatter" package (Stepner, 2013) with eight equal-sized bins for all panels.

dimensions of heterogeneity simultaneously. The effects of being less exposed to withholding and experiencing greater consolidation are similar to those in Columns 5 and 6; the effects of choosing a different collector or one of the "big two" collectors are slightly smaller in magnitude, with the latter falling just short of statistical significance, with a *p*-value of 0.110.

VII. THE CASE STUDY OF LANCASTER COUNTY

Although the four heterogeneity tests presented above demonstrate clear links between the increased collections following Act 32 and the most salient provisions of that act, they do not address the question of whether the increased collections resulted from a reduction in nonfiling (an extensive-margin response) or from better compliance by individuals who had been filing a return even prior to Act 32 (an intensive-margin response). Disentangling those channels is challenging because it requires panel data on individual taxpayers — something that simply does not exist in the public domain. In this section, however, I am able to offer some evidence on the underlying issues by examining changes in the number of tax returns filed for

Lancaster County, the sixth most populous county in the state. The reason for honing in on this county is that, unlike most counties where tax collection is conducted by for-profit entities such as Berkheimer Tax Administrator and Keystone Collections Group, which are outside the purview of the state's robust Right-to-Know Law (RTKL), collection of the EIT in Lancaster County is performed by a government entity, the Lancaster County Tax Collection Bureau (LCTCB), which falls under the purview of the RTKL.[43] In addition, unlike several government EIT collectors that came into being only after the passage of Act 32, the LCTCB had served as an EIT collector since its organization in 1959 as a joint venture of the county's school districts.[44] Therefore, it was among the only agencies in the state in possession of data for 2011 (or prior years), although changes to their tax software meant that even for this entity, only one year of data for the pre-Act 32 period was available.[45] The limited data that were obtained for Lancaster County (through filing a Right-to-Know request) nonetheless lends itself to preliminary analysis that offers interesting insights.

For the 17 school districts for which the EIT is collected by the LCTCB, I observe that the number of local EIT returns filed that report a positive income goes up by about 15 percent on average between 2011, the year immediately prior to implementation of Act 32, and 2012, the year immediately thereafter.[46] Data provided by the Pennsylvania Department of Revenue make it possible for us to also analyze the percentage change in the number of state PIT returns filed from those same school districts around this period. In sharp contrast to the large increase in the number of local EIT returns filed, the number of state PIT returns filed shows a small but insignificant *decline* between 2011 and 2012.[47] Furthermore, as evidenced by the results in the second row of Table 6, the two means are statistically different from each other at the 1 percent level.

[43] A judgment issued by an administrative officer clearly indicated that Keystone Collections Group was not an agency subject to the RTKL (https://www.openrecords.pa.gov/Appeals/DocketSheet.cfm?docket=20202119).

[44] For those reasons, the bureau served as a model agency for the state legislature to craft the Act 32 law.

[45] An open-records request I filed in 2022 with the Perry County Tax Collection District, a government entity, for the number of tax returns over time was denied by the agency on the grounds that the data did not exist. An appeal to a higher-level administrative agency, the Open Records Office, proved unsuccessful, and the Perry County Tax Collection District's denial was upheld (https://www.openrecords.pa.gov/Appeals/DocketSheet.cfm?docket=20221770).

[46] To put that 15 percent number in context, Erard and Ho (2001) estimate a nonfiling rate of 7 percent with the federal income tax in 1988, even though the IRS has had payroll withholding since 1944 and is likely far better positioned than the average local government to enforce tax compliance. Likewise, Meiselman (2018) estimates that 48 percent of Detroit's individual income tax returns in 2014 were not filed on time, even though firms located within Detroit are required to withhold and remit income taxes to the city. Falling in between those two estimates of nonfiling are estimates from Bagchi and Dušek (2021), who find an increase of 23 percent in the number of state income tax returns filed in California when the state introduced withholding of its PIT in 1971.

[47] Data on the number of local EIT and state PIT returns filed by school district are provided in Table A3.

Table 6

A Comparison of Year-over-Year Change in Local EIT Returns, Revenues, and Average Income per Return with Corresponding Changes for State PIT Returns for the 17 School Districts Served by the LCTCB

Variable	(1) Treatment Group Local EIT	(2) Control Group Pennsylvania PIT	Difference between (1) Treatment and (2) Control Groups
Percentage change in the number of local EIT and state PIT returns filed between 2011 and 2012	14.829 (5.870)	−0.409 (3.904)	15.238*** (1.898)
Percentage change in the average income reported on local EIT and state PIT returns filed between 2011 and 2012	−10.210 (3.043)	3.411 (6.849)	−13.621*** (1.820)
Percentage change in revenues collected as local EIT and state PIT between (fiscal years) 2012 and 2013	16.465 (10.242)	3.479 (4.155)	12.986*** (1.911)
Number of observations	17	17	34

Note: Data on local EIT returns were obtained through filing a Right-to-Know request with the Lancaster County Tax Collection Bureau. I appreciate the kind assistance of Chris Johnson, the executive director of the agency. Data on local EIT revenues are sourced from the Local Education Agency (School District) Finance Survey (F-33) maintained by NCES. The School District Finance Survey (F-33) data are at the fiscal year level. Accordingly, the increased collections resulting from the implementation of Act 32 on January 1, 2012, would show up primarily in the 2013 fiscal year (July 1, 2012, to June 30, 2013) for Pennsylvania. Data on state PIT returns and revenues broken out by school districts are provided by the Pennsylvania Department of Revenue through its School District Income Statistics (https://www.pa.gov/agencies/revenue/resources/reports-and-statistics/search-reports-and-statistics.html#sortCriteria=@copapwpyear%20descending&f-copapwptopic=School%20District%20Income). The comparison of mean changes in the number of returns filed, average income reported per return, and revenues collected in taxes is conducted for the 17 school districts for which EIT collection is conducted by the LCTCB. These include 16 school districts entirely contained within Lancaster County and the Octorara School District, which is split between Lancaster and Chester counties. Asterisks denote significance at the 1% (***) level.

Three other pieces of evidence also support the view that the discrepancy reported above in the growth of local EIT returns and state PIT returns relates causally to the introduction of Act 32. First, using data provided by the LCTCB, I can compute the average gross income reported per local return filed and observe that it declines by about 10 percent, from about $37,500 in 2011 to about $33,735 in 2012, when Act 32 is first implemented. Interestingly, median household income and average gross income reported per state PIT return filed both *increase* slightly for these 17 school districts, from about $58,400 in 2011 to about $59,000 in 2012 for median household income and from about $50,200 in 2011 to $51,700 in 2012 for gross income per state return.[48] These findings strengthen the view that it is changes in the composition of filers, rather than changes in household income, that explain the decreases in average gross income reported on local EIT returns. Moreover, the decline in the average gross income per return is observed only once between 2011 and 2012; in all subsequent years for which I am able to observe the average gross income reported on local tax returns and compute the year-on-year growth in that variable, I find modest increases of about 3–5 percent from one year to the next, demonstrating that the decline was unique to the first year that Act 32 came into effect. These observations are consistent with results presented in Erard and Ho (2001), who find that nonfilers have considerably lower incomes than filers, and with Holcombe and Gmeiner (2020), who find larger increases in the number of income tax returns filed lower down in the income distribution after the adoption of federal withholding in 1943.

Second, just as the state's Department of Revenue provides data on the number of PIT returns filed by residents of each school district, it also provides data on collections of the state PIT at the school district level. For the 17 school districts served by the LCTCB, revenues from the local EIT between fiscal years 2012 and 2013 jump by about 16 percent on average. In contrast, state PIT revenues for the *identical* 17 school districts exhibit a much smaller increase of about 3 percent for this time period.[49] As before, the two means are statistically different from each other at the $p < 0.001$ level, as evidenced by the results in the fourth row of Table 6. Furthermore, given that the state of Pennsylvania did not change its tax rate between 2012 and 2013, the increase in state PIT collections of 3 percent during this period is being driven by increases in the size of the tax base, increases that would mechanically drive up local income tax collections as well. Thus, after accounting for growth in the size of the underlying tax base, the impact of Act 32 on local income tax collections for the 17 districts served by the LCTCB is about $(16 - 3)$ or 13 percent, reasonably close to the 9 percent estimates I find earlier.

Third and finally, Figure 5 provides supporting evidence consistent with substantial nonfiling in the pre-Act 32 period for jurisdictions within Lancaster County. It indicates that while there was a substantial increase in the number of local income tax returns filed in 2012 when Act 32 first comes into effect, it is municipalities with *lower* per capita incomes that experience *larger* increases.[50] In contrast, there is a weak positive relationship between the change in the number of local EIT returns filed from each municipality and per capita income in the subsequent three-year period from 2013 through 2015. In other words, municipalities with lower per capita

[48] As confirmed by the results in the third row of Table 6, differences in the year-over-year change in average income reported on state versus local tax returns is statistically significant.

[49] The reader can refer to the data on local EIT and state PIT revenues collected by school district in Table A4.

[50] The Lancaster County data are broken down by school district and further by municipality. The LCTCB serves 17 school districts spread over 67 different municipalities, and I use the municipal-level data as they are more granular and offer greater statistical power.

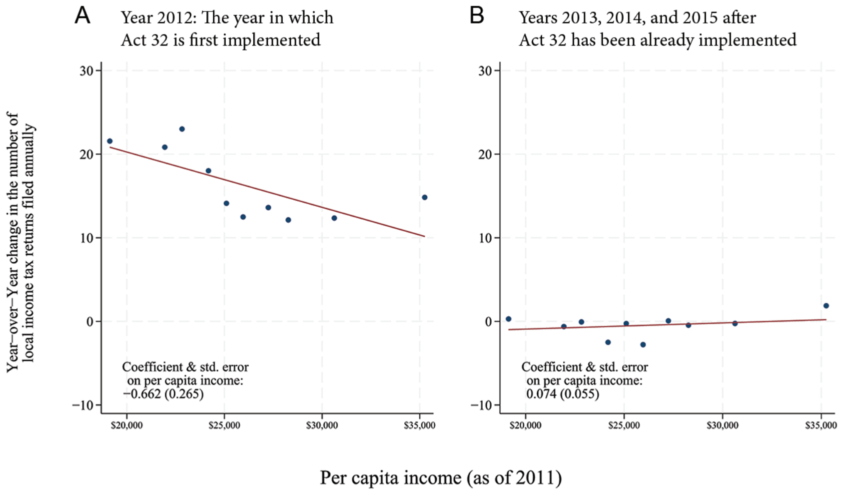

Figure 5. Scatterplot of year-over-year percentage change in the number of local EIT returns filed and its relationship with per capita income. This figure shows scatterplots of the year-over-year percentage change in the number of local EIT returns reporting a positive income against the average per capita income (measured as of 2011). The underlying data are at the municipal level and include all 67 municipalities served by the LCTCB. Data on local EIT returns were obtained through filing a Right-to-Know request with the agency. Panel A uses data for the first year in which Act 32 comes into effect (2012) and examines the percentage change in the number of returns between 2011 and 2012. Panel B examines the percentage change in the number of returns for three post-Act 32 years: between 2012 and 2013, between 2013 and 2014, and between 2014 and 2015. This figure is created using the "binscatter" package (Stepner, 2013) with 10 equal-sized bins for both panels.

incomes experienced larger increases in the number of returns filed during that first year when Act 32 was implemented, but no such relationship is evident subsequently.[51] These observations closely mimic the results in Bagchi and Dušek (2021) where the authors, using a case study for California (the second-to-last state to introduce withholding of the state PIT), find that less affluent counties experienced larger increases in the number of returns filed on average once withholding was adopted in that state in 1971. The findings of this paper go beyond those reported in Bagchi and Dušek (2021); using the ACS and state income tax data, I can show that it is changes to filing behavior following the introduction of withholding rather than changes in incomes that are driving these results.

Although the lack of data prevents me from conducting a similar analysis for other jurisdictions, there is anecdotal evidence suggesting that the decrease in

[51] I confirm these impressions through regression results (included in Table A.5) in which I find a negative (and statistically significant) relationship between the change in the number of returns filed annually and per capita income for 2012 — the first year in which Act 32 comes into effect — but a positive (and statistically insignificant) relationship for the subsequent three-year period.

nonfiling was not unique to Lancaster County. An article from Allegheny County titled “Fewer collectors, more funds” notes:[52]

> Mandatory withholding by employers has not resulted in dramatically more revenue in Mt. Lebanon, but it has meant more people are paying their earned income taxes on time. That phenomenon has been matched by a dramatic reduction in delinquent accounts. “We don’t have to chase after people,” Mt. Lebanon tax office manager Mary Abbott said.
>
> The numbers are striking. The South Hills suburb has about 20,000 wage earners. Traditionally, about 15,000 regularly mail in their quarterly tax payments on time, Ms. Abbott said. Statistics from Jordan Tax Service, the agent for Allegheny Southwest Tax Collection District of which Mt. Lebanon is a part, showed that 19,350 wage earners had paid in 2012. “That represents some 4,000 people we don’t have to pursue,” she said. . . . Ms. Abbott serves as chairwoman for Allegheny Southwest Tax Collection District, and she said she is hearing similar stories from other communities.

While it is unclear from the article as to how many of the delinquent 5,000 wage earners would have ultimately remitted their tax obligations through filing a tax return, it seems safe to conclude based on this discussion that noncompliance — stemming, in part, from nonfiling — was an issue in Allegheny County just as it was in Lancaster County. Overall, this section strongly suggests that one channel for the increased EIT revenues was a reduction in the number of nonfilers who previously had no tax liability deducted at source. Once withholding was implemented, these nonfilers came under the ambit of the tax net and faced an entirely different landscape in that the local income tax had already been withheld from their paychecks by their employers.

VIII. BROADER QUESTION: SHOULD TAX COLLECTION BE CONSOLIDATED TO AN EVEN GREATER EXTENT?

The findings that EIT revenues increased by about 9 percent following Act 32 and that the consolidation of a fragmented collection system played an important part in generating that increase raise an important question: Should EIT collection be consolidated to an even greater extent or not? The results in this paper, particularly those presented in Subsections VI.B, VI.D, and Section B of the Online Appendix, suggest that further consolidation from the present 18 collectors to a smaller number would likely result in an increase in EIT revenues, both because commission rates would go

[52] Barcousky, Len, “Less is more: Fewer tax collectors bring in more wage tax, but is it enough to lower rates,” *Pittsburgh Post Gazette*, December 12, 2013, https://www.newspapers.com/image/96440786.

down and because larger collectors have shown that they are better placed to set up effective information-sharing agreements with the state's Department of Revenue. Naturally, one could pose the question of whether EIT collection for all 2,900 jurisdictions imposing the EIT should be done centrally, similar to how the states of Iowa and Maryland collect their respective local income taxes.

Fortunately for the purposes of this discussion, we can draw on the results of a study conducted by the Pennsylvania Department of Revenue in December 2018 on the topic of "Statewide Collections Feasibility." The report acknowledges several possible improvements resulting from further consolidation, such as gaining economies of scale, making filing easier for employers, increasing standardization, and eliminating the need for employers and employees to complete multiple forms. Nonetheless, based on conversations held with numerous stakeholders including the taxing districts (counties) and the tax collectors, the report expressed some concerns that ultimately precluded them from recommending a shift to a statewide collection system, such as:

1. *Increased costs*. Taxing districts were apprehensive that a statewide collector such as the state's Department of Revenue could get by with charging higher prices and providing lower levels of service without any contractual consequences.[53] In contrast, under the current setup, taxing districts are able to select a collector once every three years or so, and should service levels be lower than what they expect, they are able to switch to a different collector. Although changing tax collectors is infrequent in practice, the mere fact that taxing districts can switch if service levels drop significantly seems to offer reassurance to the officials in charge of collection and make them reluctant to embrace a statewide filing system.
2. *Decreased cash flow*. Act 32 requires tax collectors to distribute the taxes collected within 30 days of collections from the taxpayer, with taxing districts often requiring their collectors, as part of their contracts, to turn over funds on a weekly basis. There was a concern that a statewide collector might be unable (or unwilling) to process returns and payments and to distribute funds as quickly as the current tax collectors, who also have the advantage of being more local than the state Department of Revenue based in Harrisburg.
3. *Complexities with EIT*. Unlike the states of Iowa or Maryland, where the base of the local income tax and the state personal income tax are identical, and

[53] I should note, however, that the Iowa Department of Revenue does not charge school districts any commission or fee to collect the local surtax on their behalf. This fact was confirmed in personal correspondence with John Parker on July 8, 2024. Although Maryland does not charge a commission or fee per se, state law requires the Comptroller's Office to retain a portion of the income tax revenue to cover its administrative costs, and based on the information available here, I estimate the collection costs to be about 0.4 percent of income taxes collected. This is sharply lower than the Pennsylvania average of approximately 1.5 percent.

> complying with one's local tax obligations is achieved by simply multiplying one's state tax liability (Iowa)[54] or income (Maryland)[55] with the location-specific tax rate, in Pennsylvania the local EIT and state PIT bases differ. In addition to the fact that the state taxes income from interest, dividends, and capital gains while the local EIT does not, another important difference is that the state PIT does not distinguish between passive business income (e.g., S-corporation income) and nonpassive business income, whereas the EIT base excludes passive business income. There are other smaller differences between the two tax bases, such as the exclusion of all active military pay from the local EIT but its (partial)[56] inclusion in the state PIT base as long as the taxpayer resides within the state.[57] Likewise, housing allowances for members of the clergy are included in the tax base for the state PIT but excluded from the local EIT.[58] Differences like these do make it challenging for the state PIT and the local EIT to be collected simultaneously as is done in the states of Iowa and Maryland. To the extent that policymakers would like to consolidate collection of the state and the local income taxes and have taxpayers use the same form for both, they would have to remove these differences in the tax base, even if they ultimately decided to exclude nonlabor income such as income from interest and dividends from the local EIT base while retaining them in the state PIT base.

A second question in the context of consolidation is whether consolidation should be applied to the collection of other taxes levied by local governments in Pennsylvania, perhaps at the county level similar to what Act 32 accomplished with the local EIT. At the outset, it is worth noting that Pennsylvania's tax structure is Byzantine, making it, in the words of a leading provider of payroll software, "the most complicated state in the country for local taxes."[59] Local governments in the state impose a myriad of taxes, including a property tax, the local income tax, a real estate transfer tax,[60] a local services tax, an occupation tax, a per capita tax, and a business privileges tax, among others, and the collection of these taxes is typically very fragmented.[61] A complete discussion of the possibility of consolidating the collection of all these taxes is well beyond the scope of this paper, but we

[54] See https://tax.iowa.gov/sites/default/files/2023-11/IA1040%2841001%29_print.pdf.

[55] See https://www.marylandtaxes.gov/forms/23_forms/502.pdf.

[56] See https://www.hab-inc.com/local-earned-income-tax-return-faq.

[57] See https://www.pa.gov/content/dam/copapwp-pagov/en/revenue/documents/formsandpublications/formsforindividuals/pit/documents/rev-581.pdf.

[58] See https://www.pghpresbytery.org/ministry_teams/forms__resources/business-office/financial_info_tools/questions-about-housing-allowance.

[59] See https://www.symmetry.com/local-taxes/pennsylvania-local-taxes.

[60] See https://www.alleghenycounty.us/Services/Property-Assessment-and-Real-Estate/Realty-Transfer-Taxes.

[61] See https://dced.pa.gov/local-government/local-income-tax-information/local-services-tax.

can at least consider the opportunities for consolidation presented by the property tax, the most important source of revenue for Pennsylvania local governments.

The collection of the property tax is more fragmented than it had been for the EIT even prior to Act 32. Virtually all of Pennsylvania's municipalities and school districts levy a property tax, and that collection is typically accomplished by an elected official chosen by the voters of each municipality (Bagchi, 2021). Thus, barring the few instances where a municipality has adopted a home-rule charter and abolished its elected office of the property tax collector (or designated a private firm to collect these taxes), an individual residing within the boundaries of that municipality is responsible for collecting property taxes for that municipality (along with the county and school district taxes). As a result, Pennsylvania is unique in having more than 2,400 individuals serve as property tax collectors in a state with roughly 2,600 municipalities and 500 school districts.[62] Although a 2011 report by the Pennsylvania Legislative Budget and Finance Committee (LBFC) gave serious consideration to consolidating collection at the county level, given that counties are responsible for valuing property and maintaining an inventory of all taxable property, the report ultimately backed away from that proposal, noting the opposition from groups such as the Pennsylvania State Association of Township Supervisors and the Pennsylvania State Association of Boroughs. One of the primary concerns expressed by local officials at the time of the report was that they were deeply involved in the implementation of Act 32 and anticipated devoting time and effort to that task in the subsequent months. Accordingly, they had little appetite for taking on the task of consolidating collection of the property tax as well.

However, circumstances and perspectives change, and, in a hearing conducted by the House Finance Committee in 2015 to conduct a postmortem on Act 32, an official noted, "The current system [of property tax collection is] both confusing to the taxpayer and inefficient and expensive for the government agencies."[63] He went on to suggest that in fact, Act 32 had been "working so well that the lessons learned and success achieved through EIT collections under Act 32 could be applied to the consolidation of real estate tax collections on a countywide basis." Likewise, during the LBFC's 2016 study of whether the Act 32 reforms had achieved their intended objectives, while some stakeholders did offer suggestions for the General Assembly to improve Act 32, "the most common recommendation was to use Act 32 as a model to modernize the collection of other local taxes, such as property taxes, business privilege taxes, and local service taxes." Thus, with Act 32's demonstrated success in raising collections now in the rearview mirror, one could speculate that there may be a greater interest on the part of elected officials, at both state and local levels, in consolidating collection of the property tax and other local taxes.

[62] See page 6 of the report "Pennsylvania's Current Real Property Tax Collection System."

[63] See pages 21–23 of the transcript of a hearing before the House Finance Committee on October 6, 2015 (https://www.legis.state.pa.us/WU01/LI/TR/Transcripts/2015_0132T.pdf).

IX. DISCUSSION AND CONCLUSIONS

Withholding and third-party information reporting have become important themes in public economics. While much of the existing research has focused on policies adopted by national or state governments, little attention has been paid to the experience of local governments. The introduction of withholding for local governments in Pennsylvania in 2012, following the passage of Act 32, provides me with a valuable setting to fill this gap in the literature. Using a DID research design, I find that these reforms involving the introduction of withholding and consolidation of tax collection led to an immediate and permanent increase in tax collections of about 9 percent. This is a robust result that holds up regardless of how I define the sample period around the adoption of the reforms and is not explained by increases in the size of the tax base or by changes to tax rates. Instead, the higher collections of the tax are driven by higher compliance, with the effects being larger in school districts where fewer residents had been previously subject to employer withholding and in counties that experienced a greater consolidation of tax collection. One caveat to these findings, robust as they are, is that the impact I document results from the simultaneous introduction of several policy changes, including third-party information reporting and employer remittance of withheld taxes. Thus, the findings of this paper cannot speak directly to the estimated increase in revenues that would result if only third-party reporting were introduced without also introducing the remittance of those revenues.

The increase in tax collections of 9 percent reported in this paper is not only statistically significant but also economically significant; local income taxes are the second most important source of revenue for local governments in Pennsylvania. The magnitude of this increase is also remarkable considering that employer withholding had existed for the federal income tax since 1944 and for the state PIT since 1971, and Pennsylvania residents are liable for both taxes. Thus, even as revenue agencies at other levels of government were using withholding and had access to third-party information reports from employers, the lack of withholding at the local level itself resulted in lower levels of compliance prior to 2012. The fact that withholding had existed for a small subset of employees prior to the introduction of Act 32 — those who lived and worked in the same municipality — suggests that these effects are lower-bound estimates for cases where withholding is introduced from scratch. Last, this increase in collections of 9 percent appears to be larger than what is induced by several of the interventions described in Slemrod (2019), including "letter experiments" where revenue agencies send letters that either remind taxpayers of possible penalties or, alternatively, attempt to invoke civic pride. These estimated effects are also slightly larger than the tax compliance intervention in Brockmeyer and Hernandez (2022), who find that a withholding-rate reform increased sales tax revenue by 8 percent.

All in all, the evidence presented in this paper suggests that, should governments expand the scope of withholding to activities that currently rely on self-reporting by taxpayers, revenue agencies can anticipate significant increases in taxes collected. In addition, the paper points to the salutary effects of centralized collection regimes given the evidence that the fragmented nature of tax collection prior to Act 32 was significantly associated with lower levels of tax compliance. Going forward, the Pennsylvania state legislature could explore the possibility of consolidating collection for other taxes such as the property tax and the local services tax, the collection of which remains significantly fragmented to date and perhaps the results of this paper might provide support for moves in that direction.

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DISCLOSURES

The author declares that he has no relevant or material financial interests that relate to the research described in this paper. No party had the right to review the paper prior to its circulation.

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